

NavAI

Audit & 3-Month Development Plan

13 weeks · 5 days/week · 1-2 h/day · 65 sessions
05 Oct 2026 - 01 Jan 2027

Focus: learn PostgreSQL/PostGIS, Docker and Kubernetes while turning NavAI into a correct, tested, genuinely accessibility-aware routing app. Free tiers only.

1. Executive summary

NavAI is a promising MVP: a Flask backend that fetches a Google walking route, intersects it with Madrid open data (sidewalk width, acoustic buzzers, accessibility works), asks Gemini to detect crosswalks on Street View images, and a React + Leaflet frontend that colours route segments by accessibility. The frontend lints and builds cleanly (1 minor ESLint warning), but the audit found several issues that break the app outside the author's machine or make results misleading.

- **Can't be installed elsewhere:** the Python requirements file is a Windows conda freeze, and the data lives in Git LFS (265 MB).
- **Results can be silently wrong:** CRS handling, degree-based buffers and a segment-count score (not distance-weighted).
- **The core promise isn't implemented yet:** routes are not ranked or chosen by accessibility and the impairment selector is ignored.
- **Cost/latency:** one Gemini call per route vertex, sequential, with a retired model and deprecated SDK.

The plan fixes these first (weeks 1-3), then uses the learning goals as the engine for real product improvements: **Docker** (week 4), **PostgreSQL/PostGIS** as the spatial engine (weeks 5-6), async jobs (week 7), **pgRouting** for true accessibility-aware routing (week 8), UX (week 9), **Kubernetes** (weeks 10-11), observability and community reports (week 12) and a public v1.0 release (week 13).

2. Audit findings

Checked: all Python and React sources, config, data files; ran **eslint**, **vite build** and **pyflakes**. "Fix" points to the plan week that resolves each item.

#	Severity	Area	Finding	Fix
1	Critical	Setup	<code>requirements.txt</code> is a Windows/Anaconda <code>pip freeze (@ file:///C:/... paths, pywin32, Django, Selenium, conda...)</code> . <code>pip install -r requirements.txt</code> fails on Linux/macOS and in Docker.	W1
2	Critical	Data	The 3 GeoJSON files are Git LFS pointers (sidewalks = 265 MB). Without LFS the backend catches the load error, uses empty GeoDataFrames and every segment silently becomes "Desconocida". Also <code>.gitignore</code> ignores <code>data/processed/*.geojson</code> while they are tracked.	W1
3	Critical	Geo	CRS bug in <code>src/app.py</code> : when sidewalks/works have no CRS, they are tagged EPSG:25830 but never reprojected to 4326, so intersections with the lon/lat route never match.	W1
4	High	Scoring	Global score counts polyline segments, not metres: a 2 m segment weighs the same as a 200 m one. An empty route scores 100/100.	W2
5	High	Scoring	Buffers are in degrees (<code>0.0001</code> ~ 11 m N-S but ~8.5 m E-W). Rule logic is tangled: "Unknown + buzzers" is downgraded to red (Poco accesible); buzzers are ignored when works exist; the final <code>if category == "Desconocida"</code> block is dead code.	W2
6	High	Performance	Each request intersects every segment against the full 265 MB sidewalk layer without a spatial index, then calls Gemini sequentially for every route vertex with 1280x1280 images. Slow, and burns free-tier quota.	W2, W6-7
7	High	AI	<code>google-generativeai</code> SDK is deprecated and <code>gemini-1.5-flash</code> is retired. Answer parsing is <code>"true" in text</code> ("not true" counts as true). <code>genai.configure</code> runs on every image.	W7
8	High	AI	Street View heading is fixed at 0 (north) instead of following the route; no metadata check, so locations without imagery send a grey placeholder to Gemini.	W7
9	High	Robustness	<code>get_route</code> has no timeout/status check (<code>KeyError</code> when no route). Nominatim geocoding is not bounded to Madrid ("Plaza Mayor" may resolve elsewhere), has no rate limit, and uses a new geolocator per request. Exceptions are echoed to the client.	W2

#	Severity	Area	Finding	Fix
10	Medium	Feature	Impairment selection (wheelchair / blind) is sent but ignored; only one route is computed (<code>computeAlternativeRoutes: False</code>), so nothing is actually "ranked".	W8
11	Medium	Feature	Crosswalk detections are returned by the API but never drawn on the map.	W3
12	Medium	Frontend	<code>alert()</code> with the score on mount (fires twice under StrictMode); <code>whenCreated</code> no longer exists in react-leaflet v4+/v5, so <code>mapRef</code> is never set and the map never fits the route; GeoJSON strings are <code>JSON.parsed</code> six times per render.	W3
13	Medium	Frontend	Popups read <code>feature.properties.NOMBRE</code> , which doesn't exist in buzzers/works data (fields are ID, TIPO, DISTRITO, YEAR...), so popups are always generic.	W3
14	Medium	Frontend	Backend URL hard-coded to <code>http://127.0.0.1:5000</code> ; Loading page is a fake 3 s timer and is never used.	W1, W7
15	Medium	A11y	Legend relies on colour only (not colour-blind safe) and differs from README (unknown = blue vs grey, partial = orange vs yellow). Impairment buttons lack <code>aria-pressed</code> ; no screen-reader route description for blind users.	W3, W9
16	Low	UI	Dead elements: empty navbar brand, <code>href="#"</code> links, "NavAI PRO", gear button with no action, <code>material-icons</code> class without the font, broken footer link <code>NavAI\README.md</code> , missing <code>/vite.svg</code> favicon, mixed ES/EN text, <code>lang="en"</code> .	W1, W3
17	Low	Code	<code>src/main.py</code> has a syntax error on Python < 3.12 (nested quotes in f-string); <code>main.py</code> , <code>plot.py</code> , <code>route_map.html</code> , <code>image_analysis_results.json</code> are both tracked and git-ignored; unused imports; typo <code>MOCLLOA-ARAVACA</code> in ETL.	W1
18	Low	Quality	No tests, no CI, no Docker, no <code>.env.example</code> ; README references files that don't exist (<code>etl_dataset.py</code> , <code>API_KEY.txt</code> in tree) and a placeholder live-demo link.	W2-4, W13

3. Enhancement ideas

Grouped by the value they add. Most are scheduled in the plan; the rest form a backlog.

Make the core promise real

- Accessibility-aware routing with pgRouting and per-impairment cost functions (wheelchair vs visual impairment).
- Compare "accessible" vs "fastest" route with a clear trade-off summary (extra metres vs fewer barriers).
- Distance-weighted score with a confidence value (share of the route that has no data).

Useful for people on the street

- Address autocomplete, "use my location", installable PWA with offline last route.
- Screen-reader route description and free voice guidance (Web Speech API) for blind users.
- Community obstacle reports (blocked sidewalk, broken lift, works) that expire and affect routing.

Engineering quality

- Package structure, tests (pytest, Vitest, Playwright), CI, typed API schemas, structured logs, metrics.
- Caching of geocoding, routes and AI detections to stay inside free tiers; demo mode for a never-broken live demo.

Backlog (after v1.0)

- Slope data from a DEM (critical for wheelchairs), curb cuts and tactile paving from OSM tags.
- Metro/bus stop accessibility (lifts) from CRTM open data; multimodal routes.
- TypeScript migration of the frontend; FastAPI version of the backend as a comparison exercise.
- Train a small custom crosswalk/curb-cut detector to reduce reliance on Gemini.

4. How to work the plan (and keep your GitHub graph green)

- **One issue per day.** Create the 65 issues up front (or weekly), grouped in 13 milestones. Opening issues and PRs also counts as contributions.
- **Branch -> PR -> merge the same day.** Commits only appear on your contribution graph when they reach the **default branch** (or are merged via PR), the repo is not a fork, and the commit email in `git config user.email` is verified on your GitHub account.
- **Small, real commits.** Aim for 2-4 meaningful commits per session using Conventional Commits (`feat:`, `fix:`, `test:`, `docs:...`). Avoid empty/filler commits: recruiters do read history.
- **Time-box to 1-2 h.** If a task overflows, commit what works behind a flag, note the rest in the issue, and continue next day. Friday is a natural buffer.
- **Learning notes count.** Weeks 4-5 and 10-11 include `docs/learning/*.md` notes; they are real contributions and great interview material.
- **Dates are a guide, the order is the plan.** Christmas-week sessions are marked as possible light days (docs/small tasks). On public holidays (e.g. 12 Oct, 9 Nov in Madrid, 8 Dec) or missed days, just shift the schedule.
- **Free tiers:** cache every Google/Gemini call, use kind locally for Kubernetes, GHCR for images, and free hosting for the demo (re-check limits before week 13).

Phase overview

Weeks	Phase	Main tech you'll learn
1-3	Stabilise: fixes, refactor, tests, CI, frontend	pytest, ruff, GitHub Actions, React hooks, a11y
4	Containerise	Docker, multi-stage builds, compose, GHCR
5-6	Spatial database	PostgreSQL, PostGIS, SQLAlchemy/GeoAlchemy2, Alembic, GiST
7	Async AI pipeline	google-genai, Redis, RQ workers
8	Accessible routing	pgRouting, graph algorithms, OSM
9	User experience	PWA, Web Speech API, Vitest, Playwright
10-11	Orchestration	Kubernetes (kind), StatefulSets, Ingress, Helm, HPA, CronJobs
12	Operate & harden	Prometheus, Grafana, rate limiting, CodeQL
13	Ship	Free hosting, docs, release management

5. Day-by-day plan

Week 1: Foundations & critical fixes

Mon 05 Oct - Fri 09 Oct 2026 · Make the project installable and correct on any machine.

Date	Task	What to do	Suggested commit
Mon 05 Oct	Minimal, portable dependencies	Replace the conda freeze with a hand-written <code>requirements.txt</code> (flask, flask-cors, geopandas, shapely, pyogrio, requests, geopy, googlemaps, pillow, python-dotenv) and <code>requirements-dev.txt</code> (pytest, ruff). Verify in a fresh venv.	<code>build: replace conda freeze with minimal requirements</code>
Tue 06 Oct	Config through environment	Load <code>GOOGLE_API_KEY</code> from <code>.env</code> (python-dotenv), add <code>.env.example</code> , drop <code>API_KEY.txt</code> . Frontend: <code>VITE_API_URL</code> + Vite dev proxy for <code>/api</code> .	<code>feat(config): read secrets and API URL from env</code>
Wed 07 Oct	Repository hygiene	Resolve tracked-but-ignored files (<code>main.py</code> , <code>plot.py</code> , <code>route_map.html</code> , results JSON), fix the py<3.12 f-string, remove unused imports, add favicon, fix footer README link.	<code>chore: clean tracked artefacts and small bugs</code>
Thu 08 Oct	Reproducible data pipeline	Document raw data sources (Madrid open-data URLs) in <code>data/README.md</code> ; give ETL scripts argparse paths; fix <code>MONCLOA-ARAVACA</code> typo; add <code>Makefile</code> with <code>make etl</code> .	<code>docs(data): document sources and make ETL reproducible</code>
Fri 09 Oct	Fix CRS bug + sample data	Always reproject layers to a known CRS. Create small fixture GeoJSONs (~1 km2 around Atocha) in <code>tests/fixtures</code> so the app and tests run without the 265 MB LFS file.	<code>fix(geo): reproject layers and add sample fixtures</code>

End of week 1: open a PR/merge summary, close the milestone, update the README checklist.

Week 2: Backend refactor & first tests

Mon 12 Oct - Fri 16 Oct 2026 · Turn scripts into a testable package with correct scoring.

Date	Task	What to do	Suggested commit
Mon 12 Oct	Package structure	Move backend to <code>backend/navai/</code> (<code>config</code> , <code>data</code> , <code>routing</code> , <code>scoring</code> , <code>vision</code> , <code>api</code>) with a Flask app factory <code>create_app()</code> ; add <code>/api/health</code> .	<code>refactor: split backend into navai package</code>
Tue 13 Oct	pytest + ruff	Configure <code>pyproject.toml</code> , ruff, pytest. First tests for the width thresholds of the segment classifier using fixtures.	<code>test: add pytest setup and classifier tests</code>
Wed 14 Oct	Fix the scoring model	Weight score by segment length in metres (project to EPSG:25830); empty route is an error, not 100; clean the works/buzzers rules and write them down in <code>docs/scoring.md</code> . Tests for each rule.	<code>fix(scoring): length-weighted score and explicit rules</code>
Thu 15 Oct	Metric buffers + spatial index	Use metre buffers (e.g. 8 m) in EPSG:25830 and <code>sindex.query</code> instead of full-layer <code>intersects</code> . Measure before/after timing and note it in the PR.	<code>perf(geo): metric buffers and spatial index queries</code>
Fri 16 Oct	Robust external calls	<code>get_route</code> : timeout, status check, no-route error, METRIC units. One Nominatim client with real user agent, Madrid <code>viewbox</code> , <code>RateLimiter</code> . Generic error messages to clients. Mock HTTP in tests (<code>responses</code>).	<code>fix(api): harden routing and geocoding calls</code>

End of week 2: open a PR/merge summary, close the milestone, update the README checklist.

Week 3: CI & frontend fixes

Mon 19 Oct - Fri 23 Oct 2026 · Automate quality checks and fix the map experience.

Date	Task	What to do	Suggested commit
Mon 19 Oct	GitHub Actions CI	Workflow with two jobs: backend (ruff + pytest) and frontend (eslint + vite build). Add status badge to README.	<code>ci: add backend and frontend workflows</code>
Tue 20 Oct	Map behaviour	Remove <code>alert()</code> ; show score in a panel. Replace <code>whenCreated</code> with a <code>FitBounds</code> child using <code>useMap()</code> ; parse GeoJSON once with <code>useMemo</code> .	<code>fix(map): fit route bounds and drop alert</code>
Wed 21 Oct	Real popups + crosswalks	Popups use actual fields (TIPO, DISTRITO, FECHA / YEAR, DISTRICT). Draw crosswalk detections as their own layer; label start/end markers; add a layer toggle control.	<code>feat(map): meaningful popups and crosswalk layer</code>
Thu 22 Oct	Accessible UI	Colour-blind-safe palette + dash patterns, legend consistent with README, <code>aria-pressed</code> on impairment buttons, focus styles, remove dead navbar links/gear/material-icons.	<code>feat(a11y): accessible legend and controls</code>
Fri 23 Oct	i18n EN/ES	Add react-i18next, extract all strings, language switcher, fix <code>lang</code> attribute. <code>ErrorResponse</code> : real 404 vs error, timer cleanup.	<code>feat(i18n): English and Spanish UI</code>

End of week 3: open a PR/merge summary, close the milestone, update the README checklist.

Week 4: Docker

Mon 26 Oct - Fri 30 Oct 2026 · Containerise every component. (Learning: images, layers, compose.)

Date	Task	What to do	Suggested commit
Mon 26 Oct	Backend image	Dockerfile on <code>python:3.12-slim</code> , unicorn, non-root user, layer caching for requirements. Learning notes in <code>docs/learning/docker.md</code> .	<code>build(docker): backend image</code>
Tue 27 Oct	Frontend image	Multi-stage build (node -> nginx), nginx config with SPA fallback and <code>/api</code> reverse proxy.	<code>build(docker): multi-stage frontend image</code>
Wed 28 Oct	docker compose	<code>compose.yaml</code> with frontend + backend, <code>env_file</code> , healthchecks using <code>/api/health</code> .	<code>build: docker compose stack</code>
Thu 29 Oct	Dev ergonomics	<code>compose.override.yaml</code> with bind mounts + hot reload, <code>.dockerignore</code> , <code>make up/down/logs</code> ; inspect and shrink image size.	<code>chore(docker): dev override and smaller images</code>
Fri 30 Oct	Images in CI	Build images in Actions and push to GitHub Container Registry (free) on <code>main</code> ; README "Run with Docker" section.	<code>ci: build and publish images to GHCR</code>

End of week 4: open a PR/merge summary, close the milestone, update the README checklist.

Week 5: PostgreSQL + PostGIS

Mon 02 Nov - Fri 06 Nov 2026 · Replace GeoJSON-in-memory with a spatial database. (Learning: SQL, PostGIS, migrations.)

Date	Task	What to do	Suggested commit
Mon 02 Nov	PostGIS in compose	Add <code>postgis/postgis</code> service with volume; practise <code>psql</code> , <code>ST_Transform</code> , <code>ST_DWithin</code> , <code>ST_Length</code> ; notes in <code>docs/learning/postgis.md</code> .	<code>build</code> : add PostGIS service
Tue 03 Nov	Models + migrations	SQLAlchemy 2 + GeoAlchemy2 models for sidewalks, buzzers, accessibility_works (SRID 25830). Alembic init + first migration.	<code>feat(db)</code> : spatial models and alembic migrations
Wed 04 Nov	Data loader	Idempotent loader writing processed layers into PostGIS (<code>to_postgis/ogr2ogr</code>); make <code>load-data</code> ; fixture seeding for tests.	<code>feat(etl)</code> : load layers into PostGIS
Thu 05 Nov	Spatial indexes	GiST indexes + <code>ANALYZE</code> ; compare <code>EXPLAIN ANALYZE</code> with and without index; write findings.	<code>perf(db)</code> : GiST indexes on geometry columns
Fri 06 Nov	DB in CI	PostGIS service container in GitHub Actions; integration tests run migrations + seed.	<code>ci</code> : integration tests against PostGIS

End of week 5: open a PR/merge summary, close the milestone, update the README checklist.

Week 6: Spatial logic inside PostGIS

Mon 09 Nov - Fri 13 Nov 2026 · The database does the geo work; the API gets lighter and faster.

Date	Task	What to do	Suggested commit
Mon 09 Nov	Repository layer	Query layers with <code>ST_DWithin</code> in metres instead of loading 265 MB at startup.	<code>refactor</code> : query accessibility data from PostGIS
Tue 10 Nov	One-query route analysis	Pass the whole route as a <code>LINestring</code> , split with <code>ST_DumpSegments</code> , <code>LATERAL</code> join to aggregate width/buzzers/works per segment.	<code>feat(db)</code> : single SQL route analysis
Wed 11 Nov	Parity tests	Compare SQL results with the old GeoPandas implementation on fixtures; then delete the old path.	<code>test</code> : SQL vs GeoPandas parity
Thu 12 Nov	Cache tables	<code>geocode_cache</code> , <code>route_cache</code> (hash of origin/destination/profile), <code>crosswalk_detections</code> (point, model, date). Protects the free Google quota.	<code>feat(db)</code> : caches for geocoding, routes and detections
Fri 13 Nov	Layer endpoints by bbox	<code>/api/layers/{name}?bbox=</code> using <code>ST_AsGeoJSON</code> + <code>ST_MakeEnvelope</code> ; frontend loads layers for the visible map area.	<code>feat(api)</code> : bbox-filtered layer endpoints

End of week 6: open a PR/merge summary, close the milestone, update the README checklist.

Week 7: Vision pipeline & background jobs

Mon 16 Nov - Fri 20 Nov 2026 · Modern Gemini usage, cheaper image sampling, async processing.

Date	Task	What to do	Suggested commit
Mon 16 Nov	Migrate to google-genai	Replace deprecated SDK, use a current Flash model, a single client, and structured JSON output (pydantic schema with confidence) instead of substring matching.	<code>feat(vision): migrate to google-genai structured output</code>
Tue 17 Nov	Smarter Street View	Call the free metadata endpoint first to skip places without imagery; heading from route bearing; 640x640 images.	<code>feat(vision): heading-aware Street View with metadata check</code>
Wed 18 Nov	Sampling + cache	Sample points every ~30 m (<code>ST_LineInterpolatePoints</code>) instead of every vertex; cap images per request; read detection cache first.	<code>perf(vision): distance-based sampling with cache</code>
Thu 19 Nov	Job queue	Redis + RQ worker in compose; <code>POST /api/routes</code> returns <code>job_id</code> ; <code>GET /api/jobs/{id}</code> for status/result.	<code>feat(api): async route analysis jobs</code>
Fri 20 Nov	Real loading page	Frontend polls job status with progress and retry/error states; delete the fake 3 s timer.	<code>feat(ui): progress-driven loading page</code>

End of week 7: open a PR/merge summary, close the milestone, update the README checklist.

Week 8: Accessibility-aware routing (pgRouting)

Mon 23 Nov - Fri 27 Nov 2026 · Make "accessible route" real: routes chosen by accessibility, per impairment.

Date	Task	What to do	Suggested commit
Mon 23 Nov	Pedestrian graph	Switch to a pgRouting-enabled image; import the OSM pedestrian network for central Madrid (<code>osmnx</code> or <code>osm2pgrouting</code>) into an <code>edges</code> table.	<code>feat(routing): import pedestrian graph</code>
Tue 24 Nov	Enrich edges	Migration/SQL that joins sidewalk width, buzzers and works onto edges.	<code>feat(routing): accessibility attributes on edges</code>
Wed 25 Nov	Profile cost functions	Wheelchair: heavy penalty/block for width < 0.9 m; visual: prefer crossings with buzzers. <code>pgr_dijkstra</code> with cost expressions; unit tests on a tiny graph.	<code>feat(routing): impairment-specific cost functions</code>
Thu 26 Nov	Routes endpoint	<code>/api/routes?profile=</code> returns the accessible route plus the Google fastest route for comparison. Impairment selector now matters.	<code>feat(api): profile-based accessible routing</code>
Fri 27 Nov	Compare in UI	Show both routes, toggle, summary card (distance, time, score, % unknown data).	<code>feat(ui): accessible vs fastest comparison</code>

End of week 8: open a PR/merge summary, close the milestone, update the README checklist.

Week 9: User experience

Mon 30 Nov - Fri 04 Dec 2026 · Features that make the app genuinely useful on the street.

Date	Task	What to do	Suggested commit
Mon 30 Nov	Autocomplete + my location	Debounce address search (Photon/Nominatim, Madrid-bounded) and "use my location" via Geolocation API.	<code>feat(ui): address autocomplete and geolocation</code>
Tue 01 Dec	Route details panel	Segment list with issues; plain-text route description for screen readers with <code>aria-live</code> .	<code>feat(a11y): screen-reader route description</code>
Wed 02 Dec	Voice guidance	Read the description with the browser Web Speech API (free, replaces paid Cloud TTS).	<code>feat(a11y): voice route guidance</code>
Thu 03 Dec	PWA + mobile	vite-plugin-pwa manifest and service worker, offline cache of last route, responsive layout.	<code>feat(pwa): installable offline-capable app</code>
Fri 04 Dec	Frontend tests	Vitest + React Testing Library for Home/Navigation; one Playwright smoke test with mocked API in CI.	<code>test(ui): unit and e2e smoke tests</code>

End of week 9: open a PR/merge summary, close the milestone, update the README checklist.

Week 10: Kubernetes fundamentals (kind)

Mon 07 Dec - Fri 11 Dec 2026 · Run the whole stack on a local, free cluster. (Learning: pods, services, config, storage.)

Date	Task	What to do	Suggested commit
Mon 07 Dec	First deployment	Install kind + kubectl; cluster config in <code>k8s/</code> ; backend Deployment + Service from the GHCR image; notes in <code>docs/learning/kubernetes.md</code> .	<code>feat(k8s): backend deployment on kind</code>
Tue 08 Dec	Config & secrets	ConfigMap + Secret (API key), frontend Deployment + Service, test with <code>port-forward</code> .	<code>feat(k8s): frontend, configmap and secrets</code>
Wed 09 Dec	Stateful Postgres	PostGIS StatefulSet with PVC + headless Service; migration/seed Kubernetes Job.	<code>feat(k8s): PostGIS statefulset and migration job</code>
Thu 10 Dec	Ingress + probes	ingress-nginx in kind: <code>/</code> -> frontend, <code>/api</code> -> backend; readiness/liveness probes on <code>/api/health</code> .	<code>feat(k8s): ingress and health probes</code>
Fri 11 Dec	Workers & resources	Redis + worker Deployments; requests/limits; <code>make k8s-up</code> script; docs.	<code>feat(k8s): redis, worker and resource limits</code>

End of week 10: open a PR/merge summary, close the milestone, update the README checklist.

Week 11: Kubernetes advanced

Mon 14 Dec - Fri 18 Dec 2026 · Package, scale and automate.

Date	Task	What to do	Suggested commit
Mon 14 Dec	Helm chart (1/2)	<code>helm create navai</code> ; template deployments/services from values.	<code>feat(helm): initial chart</code>
Tue 15 Dec	Helm chart (2/2)	Values for images, replicas, secrets; <code>helm lint</code> in CI; install on kind.	<code>feat(helm): configurable values and lint in CI</code>
Wed 16 Dec	Autoscaling	metrics-server + HPA for backend; k6 load test (free) and screenshot the scaling.	<code>feat(k8s): HPA with load test</code>
Thu 17 Dec	ETL CronJob	Weekly CronJob that downloads Madrid open data, transforms and upserts into PostGIS.	<code>feat(k8s): scheduled open-data refresh</code>
Fri 18 Dec	Cluster e2e in CI	Actions job: create kind cluster, <code>helm install</code> , curl smoke test.	<code>ci: kind-based end-to-end test</code>

End of week 11: open a PR/merge summary, close the milestone, update the README checklist.

Week 12: Observability, community & security

Mon 21 Dec - Fri 25 Dec 2026 · Know what's happening; let users contribute; harden.

Date	Task	What to do	Suggested commit
Mon 21 Dec	Logs + metrics	Structured JSON logs with request IDs; Prometheus metrics (latency, Gemini calls, cache hit ratio).	<code>feat(obs): structured logs and prometheus metrics</code>
Tue 22 Dec	Dashboards	kube-prometheus-stack on kind; commit Grafana dashboard JSON.	<code>feat(obs): grafana dashboard</code>
Wed 23 Dec	Obstacle reports API	Table <code>obstacle_reports</code> (point, type, description, expires_at); validated POST/GET endpoints.	<code>feat(api): community obstacle reports</code>
Thu 24 Dec	Reports in the map/routing	Click-to-report form; show active reports; reports add routing cost. (Light day if holiday.)	<code>feat(ui): report obstacles from map</code>
Fri 25 Dec	Security pass	Restrict CORS, request size limits, flask-limiter, Dependabot, CodeQL; fix findings. (Light day if holiday.)	<code>chore(security): hardening and dependency scanning</code>

End of week 12: open a PR/merge summary, close the milestone, update the README checklist.

Week 13: Deploy & release v1.0

Mon 28 Dec - Fri 01 Jan 2027 · Public demo and a portfolio-ready repository.

Date	Task	What to do	Suggested commit
Mon 28 Dec	Free deployment	Frontend on Vercel/Netlify/GitHub Pages; backend on a free container host; free Postgres with PostGIS (e.g. Neon/Supabase). Check current free-tier limits first. Optional: k3s on Oracle Cloud Always Free.	<code>chore(deploy): public demo environment</code>
Tue 29 Dec	Demo mode	Pre-seeded sample routes that work with no API key / exhausted quota, so the live demo never breaks.	<code>feat: demo mode with seeded routes</code>
Wed 30 Dec	README rewrite	Mermaid architecture diagram, Docker/K8s instructions, GIF, real live-demo link.	<code>docs: rewrite README for v1</code>
Thu 31 Dec	Architecture decisions	ADRs in <code>docs/adr/</code> (PostGIS, pgRouting, RQ, Helm); refresh Power BI source to PostGIS or CSV export. (Light day if holiday.)	<code>docs: architecture decision records</code>
Fri 01 Jan	Release v1.0.0	CHANGELOG, tag, GitHub Release, retrospective + LinkedIn/blog post, backlog for next quarter. (Light day if holiday.)	<code>release: v1.0.0</code>

End of week 13: open a PR/merge summary, close the milestone, update the README checklist.

6. Progress tracker

Tick each session when its PR is merged.

Week	Mon	Tue	Wed	Thu	Fri	Milestone
1	☐	☐	☐	☐	☐	Foundations & critical fixes
2	☐	☐	☐	☐	☐	Backend refactor & first tests
3	☐	☐	☐	☐	☐	CI & frontend fixes
4	☐	☐	☐	☐	☐	Docker
5	☐	☐	☐	☐	☐	PostgreSQL + PostGIS
6	☐	☐	☐	☐	☐	Spatial logic inside PostGIS
7	☐	☐	☐	☐	☐	Vision pipeline & background jobs
8	☐	☐	☐	☐	☐	Accessibility-aware routing (pgRouting)
9	☐	☐	☐	☐	☐	User experience
10	☐	☐	☐	☐	☐	Kubernetes fundamentals (kind)
11	☐	☐	☐	☐	☐	Kubernetes advanced
12	☐	☐	☐	☐	☐	Observability, community & security
13	☐	☐	☐	☐	☐	Deploy & release v1.0

Definition of done for every session

- Code merged to the default branch through a PR linked to the day's issue.
- CI green (from week 3 onward); new logic has at least one test.
- README/docs updated when behaviour or setup changes.
- No secrets committed; `.env.example` updated when a new variable appears.